

Financial Performance Analytics Project Workflow

Project Objective

Develop an end-to-end Financial Analytics solution that extracts company financial data from the Alpha Vantage API, transforms it using Python, and presents interactive Power BI dashboards for financial analysis and company comparison.

Workflow

Alpha Vantage API → Python (Requests) → JSON Extraction → Pandas Cleaning → KPI Calculation → Excel Dataset → Power BI Model → DAX Measures → Interactive Dashboards

Python Modules Used

requests, pandas

Data Sources

Company Overview, Income Statement, Balance Sheet, Cash Flow Statement

Financial KPIs

Revenue Growth, Net Income Growth, Gross Margin, Profit Margin, Debt Ratio, Equity Ratio, Cash Ratio, Free Cash Flow, Cash Conversion Ratio

Dashboards

Apple Financial Performance Dashboard (3 pages); Multi-Company Financial Performance Dashboard (2 pages)

Tools

Python, REST API, Pandas, Excel, Power BI, DAX